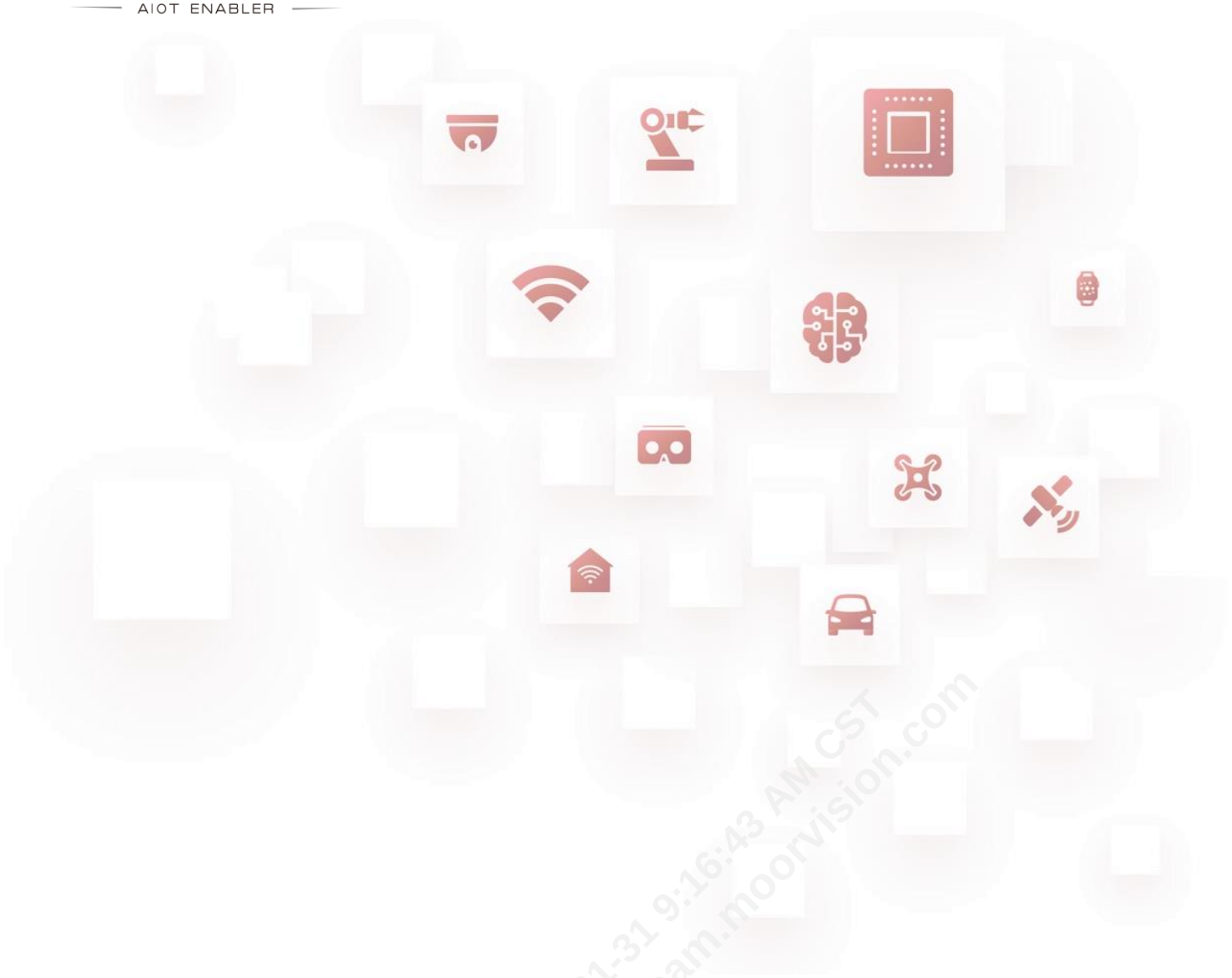
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Thundercomm TurboX C610 Open Kit UVC/UAC Device for LE2.0

User Manual

Rev. V1.0
Mar 05, 2024

DN: tc-g2-231510

Revision History

Revision	Date	Author	Description
1.0	Mar 05, 2024	Ya'nan, Li	Initial release.

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About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders, path and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Overview

This document provides instructions for using the UVC/UAC device on TurboX C610 LE2.0 baseline. The TurboX C610 open kit board can be utilized as a USB-UVC camera to transmit data to a PC using the UVC/UAC device communication protocol. In this solution, the PC operates as the host, while the C610 open kit board functions as the device.

The overall framework is shown as follows:

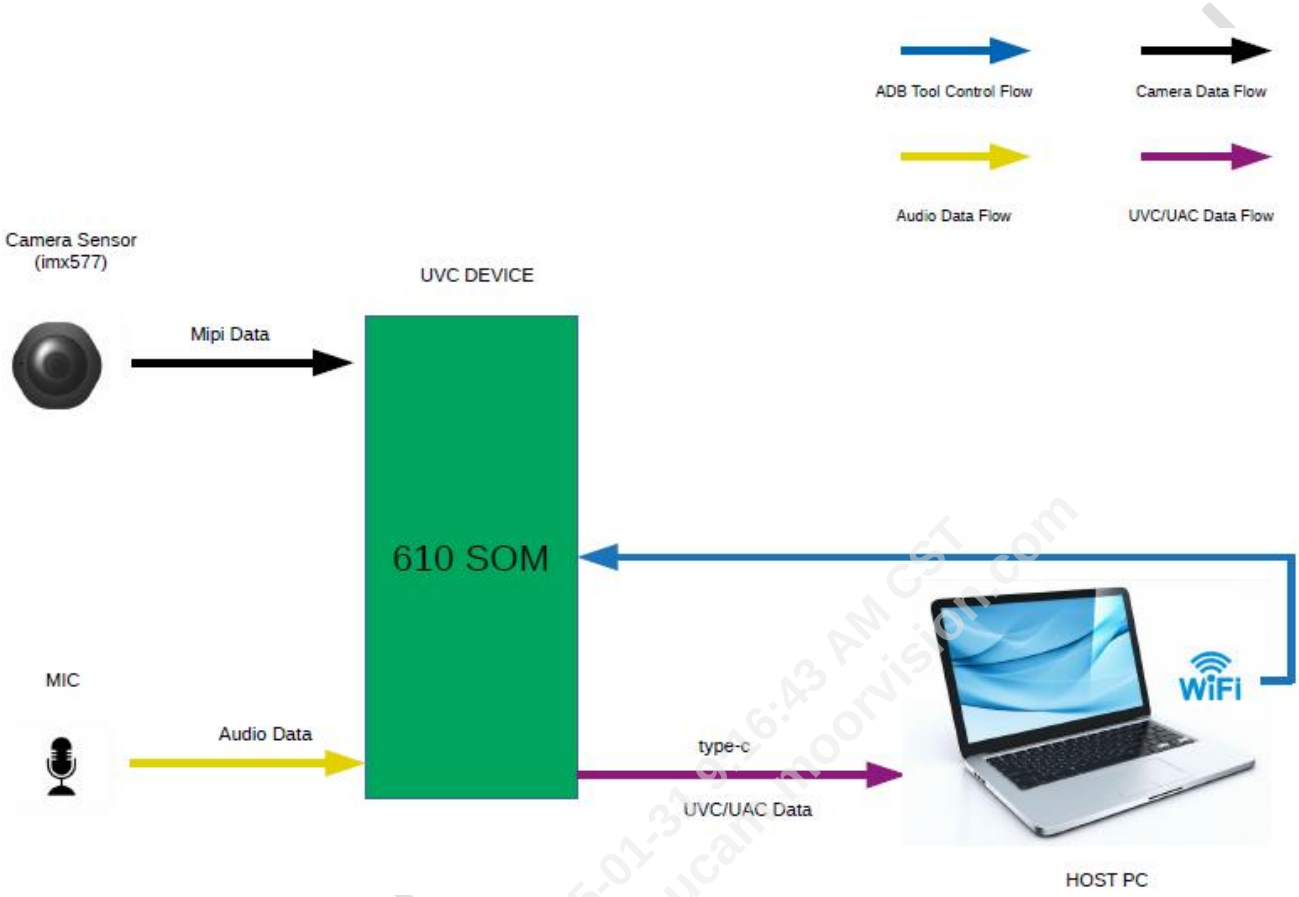


Figure 1-1. Overall Framework

➲ NOTES:

- "TurboX" referred to herein is the English text of our registered trademark: TURBOX.
- This solution is a customized implementation and is not included in the standard C610 LE2.0 SDK.

Chapter 2. Preconditions

2.1. Prepare test equipment

Step 1. Plug in the IMX577 camera.

Step 2. Plug in an LCD monitor.

➲ **NOTE:** If the LCD screen is not needed, skip this step.

Step 3. Prepare a Windows PC.

Step 4. Create a FlatBuild package using the SDK by referring to sections 3.4 to 3.5 of the file provided. Alternatively, download the FlatBuild package from section 3.6 and flash it onto the board.

TurboX_C410_C610_Open_Kit_LE_Software_Release_Notes_(for LE2.0.Post-CS1.r002002).pdf

Please refer to the document provided for using the SDK to create a Flatbuild package, or download the Flatbuild package from Chapter 3.6 of the document and flash it onto the board.

2.2. Install MPC-HC on Windows

Step 1. Please download the exe installation file of MPC-HC from the following website:

https://github.com/clsid2/mpc-hc/releases/tag/2.0.0

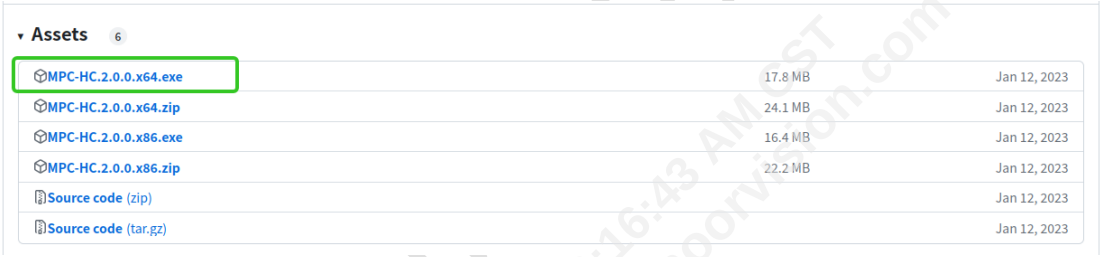


Figure 2-1. MPC-HC Installation File

Step 2. Run the *MPC-HC.2.0.0.x64.exe* file.

Step 3. Click the **Next** button.

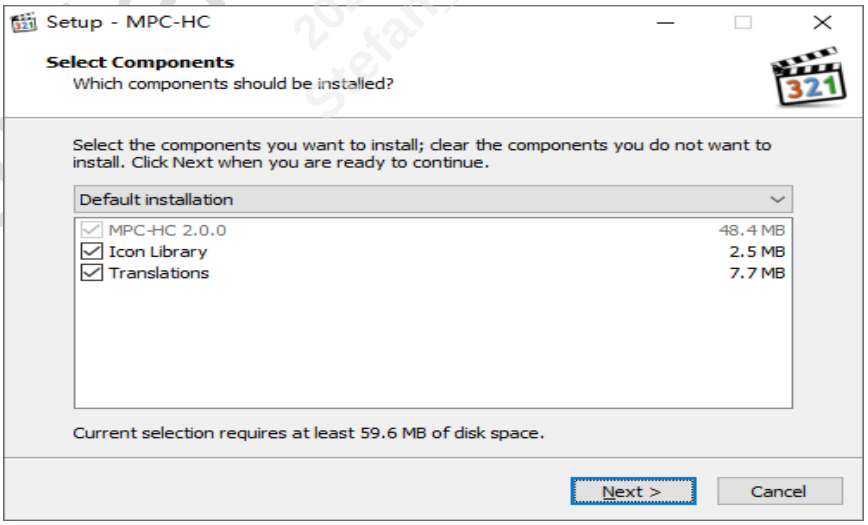


Figure 2-2. Set Up MPC-HC

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Step 4. Click **Next** to continue.

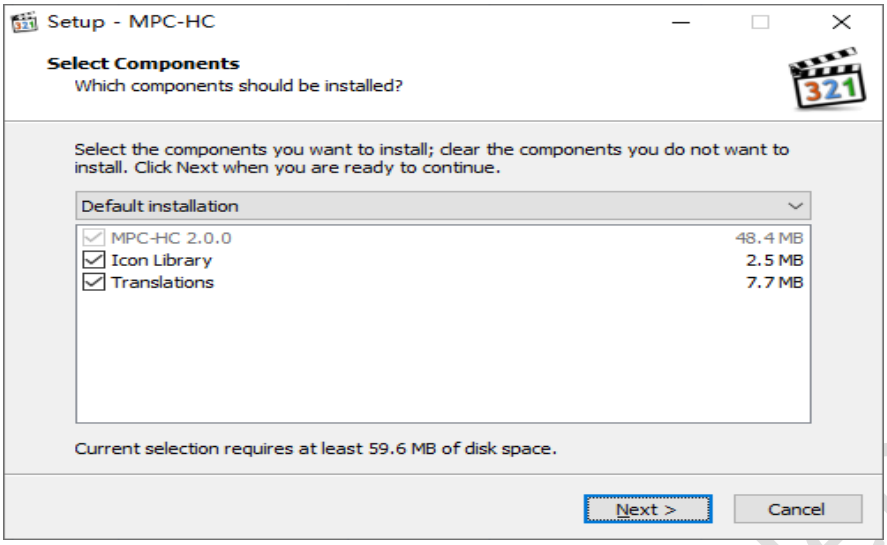


Figure 2-3. Select Components

Step 5. Click the **Install** button.

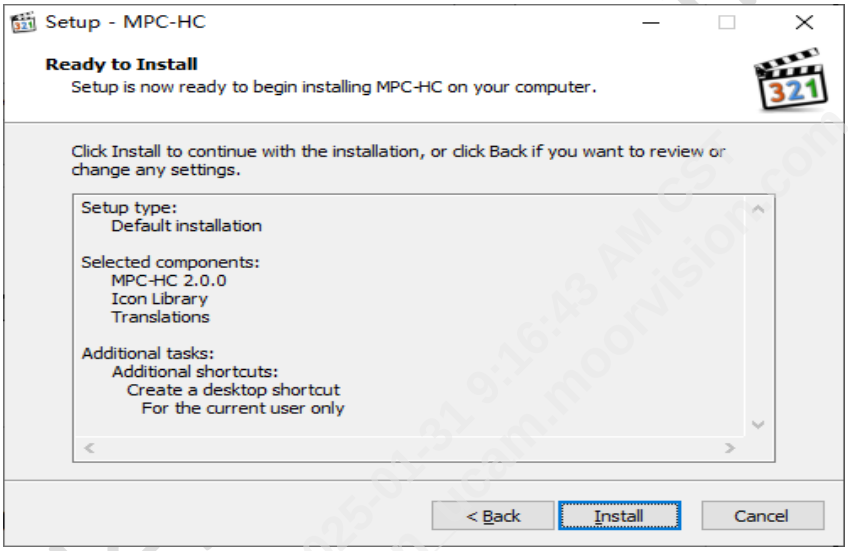


Figure 2-4. Install MPC-HC

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Step 6. Click the **Finish** button.

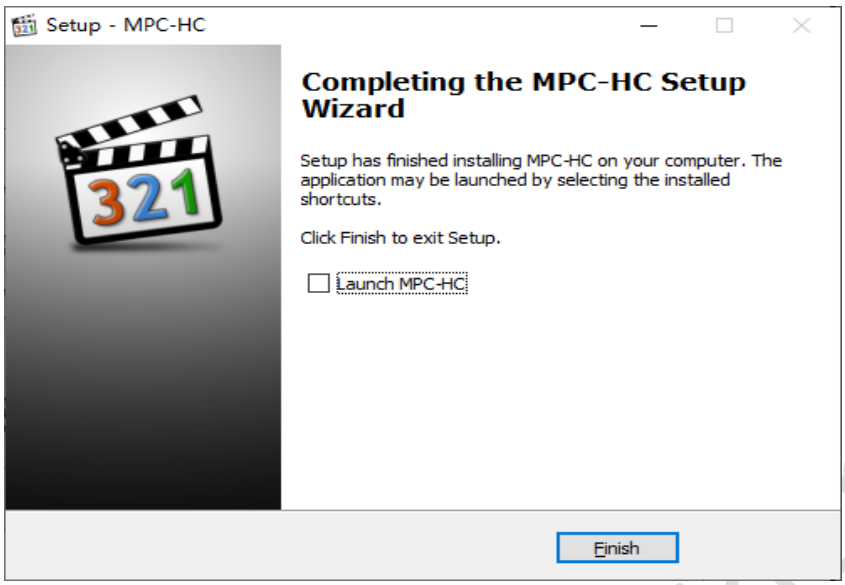


Figure 2-5. Complete MPC-HC Setup

2.3. Import and configure test files

Step 1. Disable verity.

```
$ adb root
$ adb disable-verity
$ adb reboot
```

Step 2. After rebooting the board, push the required test files onto the board.

```
$ adb root
$ adb remount
$ adb push yolov5m-320x320-int8.tflite /data/
$ adb push yolov5m.labels /data/
$ adb shell
# echo outputFormat=0 > /etc/camera/camxoverridesettings.txt
# sync
```

➲ **NOTES:**

- If the prompt "`remote Read-only file system`" appears, you will need to execute the following command to successfully perform the copy operation:

  `adb shell mount -o remount,rw /`

- You can download the two Yolov5 files mentioned in the above commands via the link below:

  Yolov5 files for 610-LE-2-0 UVC.zip

Step 3. Modify the */etc/xdg/weston/weston.ini* file using the vi editor.

```
$ adb root
$ adb shell
# vi /etc/xdg/weston/weston.ini
```

Step 4. Apply the following modification in the vi editor:

```
[output]
name=Virtual-1
mode=off
[output]
name=DSI-1
mode=on
[output]
name=DP-1
mode=off
```

Step 5. Save the modifications.

```
# sync
```

Step 6. After completing the above steps, you can proceed with the following test without restarting the board.

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Chapter 3. Start UVC and UAC

3.1. Connect the board to Wi-Fi

Step 1. Restart the *wpa_supplicant*.

```
$ adb root
$ adb shell
# killall wpa_supplicant
# wpa_supplicant -iwlan0 -Dnl80211 -c /etc/misc/wifi/wpa_supplicant.conf &
```

Step 2. Upon completion of command execution, press the **Enter** key twice.

```
liyn@liyn-HP-EliteDesk-880-G6-Tower-PC:~$ adb root
restarting adbd as root
adb shell
liyn@liyn-HP-EliteDesk-880-G6-Tower-PC:~$ adb shell
/ # killall wpa_supplicant
/ # wpa_supplicant -iwlan0 -Dnl80211 -c /etc/misc/wifi/wpa_supplicant.conf  &
/ # Successfully initialized wpa_supplicant

/ #
```

Figure 3-1. Command Execution

Step 3. Open the **wpa_cli** tool.

```
# wpa_cli -iwlan0
```

Enter the interactive mode as follows:

```
/ # wpa_cli -iwlan0
wpa_cli v2.10-devel
Copyright (c) 2004-2019, Jouni Malinen <j@w1.fi> and contributors

This software may be distributed under the terms of the BSD license.
See README for more details.


Interactive mode

>
```

Figure 3-2. Enter Interactive Mode

Step 4. Enter the Wi-Fi name (SSID) and password.

```
> add_network
> set_network 1 ssid "TP-LINK_4295"
> set_network 1 psk "12345678"
> enable_network 1
```

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In the commands, `ssid` is followed by the Wi-Fi name, and `psk` is followed by the password.

```
/ # wpa_cli -iwlan0
wpa_cli v2.10-devel
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See README for more details.



Interactive mode

> add_network
1
> set_network 1 ssid "TP-LINK_4295"
OK
> set_network 1 psk "12345678"
OK
> enable_network 1
OK
<3>CTRL-EVENT-SCAN-STARTED
> wlan0: Trying to associate with SSID 'TP-LINK_4295'
<3>CTRL-EVENT-SCAN-RESULTS
<3>WPS-AP-AVAILABLE
<3>Trying to associate with SSID 'TP-LINK_4295'
> wlan0: Associated with 58:41:20:b1:42:95
wlan0: CTRL-EVENT-SUBNET-STATUS-UPDATE status=0
<3>Associated with 58:41:20:b1:42:95
<3>CTRL-EVENT-SUBNET-STATUS-UPDATE status=0
> wlan0: WPA: RX message 1 of 4-Way Handshake from 58:41:20:b1:42:95 (ver=2)
wlan0: WPA: Sending EAPOL-Key 2/4
<3>WPA: RX message 1 of 4-Way Handshake from 58:41:20:b1:42:95 (ver=2)
<3>WPA: Sending EAPOL-Key 2/4
> wlan0: WPA: RX message 3 of 4-Way Handshake from 58:41:20:b1:42:95 (ver=2)
wlan0: WPA: Sending EAPOL-Key 4/4
wlan0: WPA: Key negotiation completed with 58:41:20:b1:42:95 [PTK=CCMP GTK=CCMP]
wlan0: CTRL-EVENT-CONNECTED - Connection to 58:41:20:b1:42:95 completed [id=1 id_str=]
<3>WPA: RX message 3 of 4-Way Handshake from 58:41:20:b1:42:95 (ver=2)
<3>WPA: Sending EAPOL-Key 4/4
<3>WPA: Key negotiation completed with 58:41:20:b1:42:95 [PTK=CCMP GTK=CCMP]
<3>CTRL-EVENT-CONNECTED - Connection to 58:41:20:b1:42:95 completed [id=1 id_str=]
>
>
>
>
```

Figure 3-3. Enter Wi-Fi Name (SSID) and Password

Step 5. Obtain an IP address from the DHCP server and configure network settings automatically.

```
# dhcpcd
```

The following output indicates that the Wi-Fi connection has been successfully established:

```
/ # dhcpcd
dhcpcd[1247]: version 5.2.10 starting
dhcpcd[1247]: gretap0: up_interface: Cannot assign requested address
dhcpcd[1247]: erspan0: up_interface: Cannot assign requested address
dhcpcd[1247]: p2p0: up_interface: Operation not supported
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
chmod: /etc/resolv.conf: Operation not permitted
dhcpcd[1247]: dummy0: broadcasting for a lease
dhcpcd[1247]: wlan0: broadcasting for a lease
dhcpcd[1247]: eth0: waiting for carrier
dhcpcd[1247]: gretap0: waiting for carrier
dhcpcd[1247]: erspan0: waiting for carrier
dhcpcd[1247]: bridge0: waiting for carrier
dhcpcd[1247]: wlan1: waiting for carrier
dhcpcd[1247]: p2p0: waiting for carrier
dhcpcd[1247]: wlan0: offered 192.168.1.112 from 192.168.1.1
dhcpcd[1247]: wlan0: acknowledged 192.168.1.112 from 192.168.1.1
dhcpcd[1247]: wlan0: checking for 192.168.1.112
dhcpcd[1247]: wlan0: leased 192.168.1.112 for 7200 seconds
chmod: /etc/resolv.conf: Operation not permitted
dhcpcd[1247]: forked to background, child pid 1352
/ #
```

Figure 3-4. Wi-Fi Connection Established

➲ **NOTE:** Please take note of the above IP address as it will be used to establish a network connection with the board.

3.2. Connect to the board via Wi-Fi

Step 1. Open a new terminal and execute the command below to switch the adb mode from Type-C to Wi-Fi:

```
$ adb tcpip 5556
```

```
liyn@liyn-HP-EliteDesk-880-G6-Tower-PC:~$ adb tcpip 5556
restarting in TCP mode port: 5556
liyn@liyn-HP-EliteDesk-880-G6-Tower-PC:~$
```

Figure 3-5. Switch from Type-C to Wi-Fi

Step 2. Establish a network connection with the board using Wi-Fi.

```
$ adb connect 192.168.1.112:5556
$ adb -s 192.168.1.112:5556  shell
```

```
liyn@liyn-HP-EliteDesk-880-G6-Tower-PC:~$ adb tcpip 5556
restarting in TCP mode port: 5556
liyn@liyn-HP-EliteDesk-880-G6-Tower-PC:~$ adb connect 192.168.1.112:5556
connected to 192.168.1.112:5556
liyn@liyn-HP-EliteDesk-880-G6-Tower-PC:~$
liyn@liyn-HP-EliteDesk-880-G6-Tower-PC:~$ adb -s 192.168.1.112:5556  shell
/ #
```

Figure 3-6. Establish Network Connection

As shown in the above figure, once the connection is established, the DK board will be controlled via Wi-Fi instead of the Type-C interface. The Type-C interface will be exclusively used for transmitting UVC and VAC data.

3.3. Open Weston

Execute the following commands to open Weston:

```
# setenforce 0
# export WESTON_DISABLE_ATOMIC=1
# export XDG_RUNTIME_DIR=/run/user/root
# weston --tty=1 --idle-time=123456 &
```

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3.4. Switch the mode of USB Type-C

Switch the mode from USB Type-C to UVC/UAC.

Step 1. Open a new terminal window.

Step 2. Establish a connection to the device with the following command:

```
$ adb -s 192.168.1.112:5556 shell
```

Step 3. Execute the following command:

```
# usb_composition
```

Step 4. Enter "9113", "n", "y", "y", and "y" in sequence and press **Enter** twice.

```
   90CA -       DIAG + ADB + UAC2
   90CB -       DIAG + ADB + UVC
   90CC -       DIAG + ADB + UAC2 + UVC
   90CD -       DIAG + ADB + GNSS
   90CE -       CCID
   90D1 -       CCID + DIAG + ADB
   90D5 -       DIAG + ADB + MBIM + GNSS + DUN
   90D6 -       DIAG + MBIM + GNSS + DUN
   90DB -       DIAG + DUN + RMNET + DPL + QDSS(Trace) + ADB
   90DF -       DIAG + UVC
   90E0 -       DIAG + UAC2 + UVC
   90E2 -       MBIM + GNSS
   90E5 -       DIAG MSM + DIAG_MDM + QDSS MSM + QDSS_MDM + DUN + DPL + RMNET + ADB
   90E7 -       RNDIS + DIAG MSM + DIAG_MDM + QDSS MSM + QDSS_MDM + DUN + DPL + ADB
   90EC -       DIAG + ADB + MBIM + GNSS + DUN + DPL
   90EF -       RNDIS + RmNet + IPC_ROUTER + DIAG
   90F0 -       RNDIS + RmNet + IPC_ROUTER + DIAG + MODEM + NMEA + ADB
   90F2 -       DIAG + ADB + DUN + DPL
   90FC -       DIAG + DUN + Muliplexed 15 pdn RMNET + DPL + QDSS MSM DATA + ADB
   90FD -       DIAG + IPC_ROUTER + RmNet + RmNet + ECM (s/w path)
   9102 -       DIAG + IPC_ROUTER + RmNet + RmNet + ECM + QDSS (Trace Only) + ADPL + ADB
   9103 -       DIAG + IPC_ROUTER + RmNet + RmNet + ECM + ADPL
   9104 -       DIAG + QDSS (Trace Only) + ADPL + ADB
   9105 -       DIAG + ADPL
   9106 -       DIAG + IPC_ROUTER + RmNet + ECM + QDSS (Trace Only) + ADPL + ADB
   9107 -       DIAG + IPC_ROUTER + RmNet + ECM + ADPL
   9108 -       DIAG + UAC1 + ADB
   9109 -       DIAG + UAC1
   9113 -       DIAG + UAC1 + UVC + ADB
   9114 -       DIAG + UAC1 + UVC
   9118 -       DIAG + UAC1 + HID
   9119 -       DIAG + UAC1 + HID + ADB
   9120 -       DIAG + UAC1 + UVC + HID
   9121 -       DIAG + UAC1 + UVC + HID + ADB
   F000 -       Mass Storage
   F005 -       ACM
   empty -      it is used to allow either hsic or hsusb to have no composition at all(must reboot to take effect).
   hsic_next -
   hsusb_next -
Pid number : 9113
Choose core: y - hsic, n - hsusb  ? (y/n)n
Would you like it to be the default composition ? (y/n)y
Would you like the composition to change immediately? (y/n)y
Are you performing the composition switch from adbd? (y/n)y
cat: can't open '/sys/bus/platform/devices/qcom_gadget/enabled': No such file or directory
/ # binding UDC with Gadget... a600000.dwc3

/ #
/ #
/ #
/ #
/ #
```

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Figure 3-7. USB Composition

➲ **NOTE:** The prompt inside the red box can be ignored.

3.5. Switch to UVC device mode

Step 1. Execute the following commands to switch the board to UVC device mode.

```
# source /etc/profile && chown -R qmmfsvr:qmmfsvr /run/user/root
# gst-umd-daemon -v /dev/video3 -a hw:1,0
```

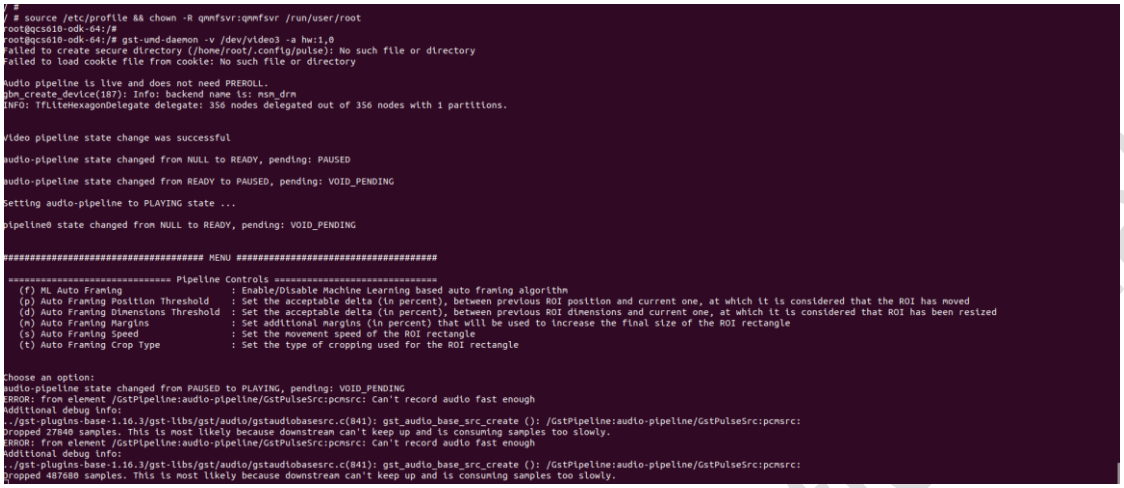


Figure 3-8.Switch to UVC Device Mode

Step 2. Connect the board to PC with MPC-HC installed using a Type-C cable, then you will be able to receive video or audio data through MPC-HC.

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Chapter 4. Open UVC and UAC Streams

Step 1. Open the **MPC-HC** software on Windows and click **View** -> **Options**.

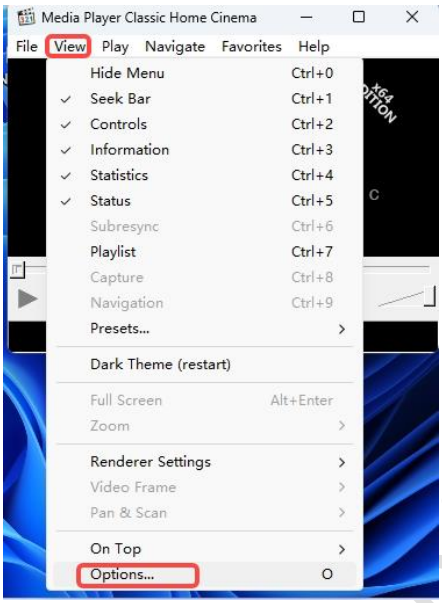


Figure 4-1. Open MPC-HC

Step 2. Click on the "**Capture**" option, select "**UVC Camera**" as the video device, and confirm your selection by clicking the confirmation button.

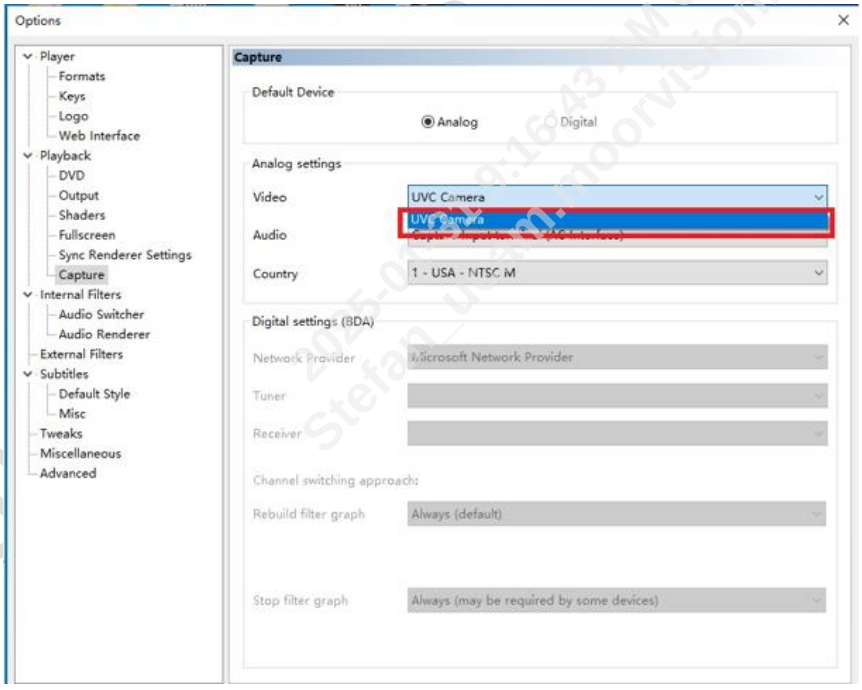


Figure 4-2. Select Video - UVC Camera

Step 3. Click on the "**Capture**" option, select "**Capture Input terminal (AC Interface)**" as the audio device, and confirm your selection by clicking the confirmation button.

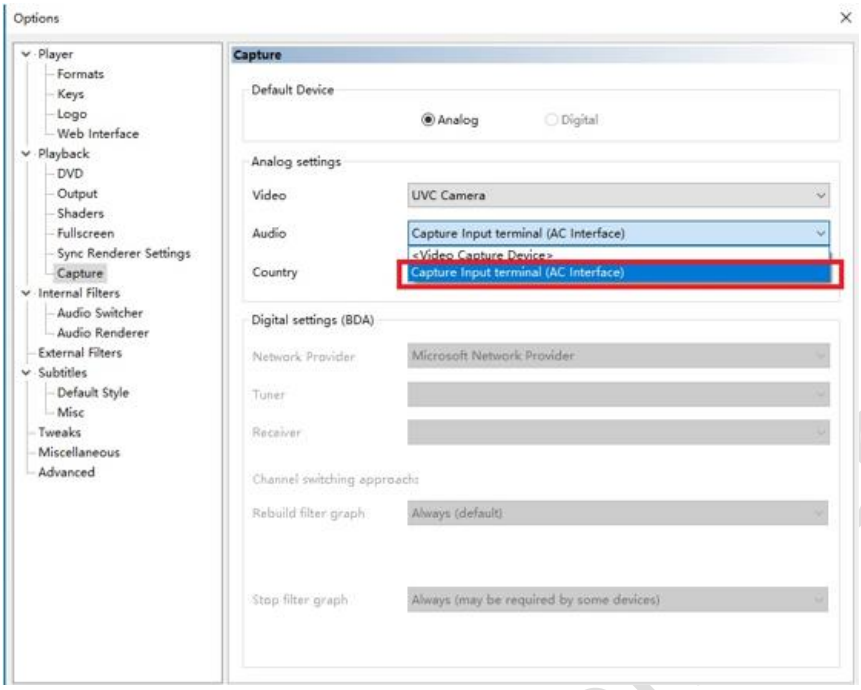


Figure 4-3. Select Video - Thundercomm (Audio)

Step 4. Select "**File**" -> "**Open Device**".

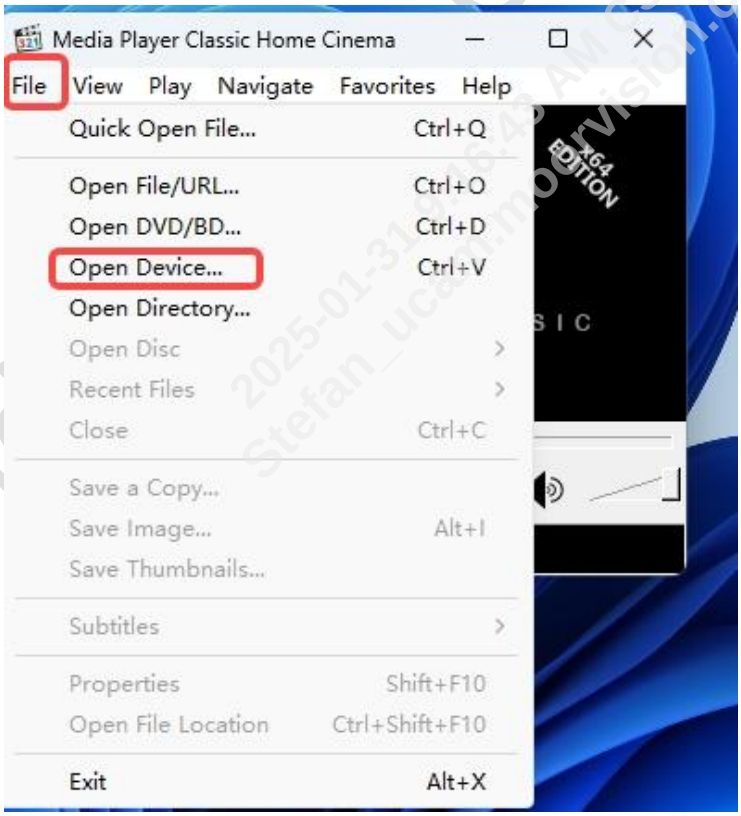


Figure 4-4. Open Device

Step 5.Change the format.

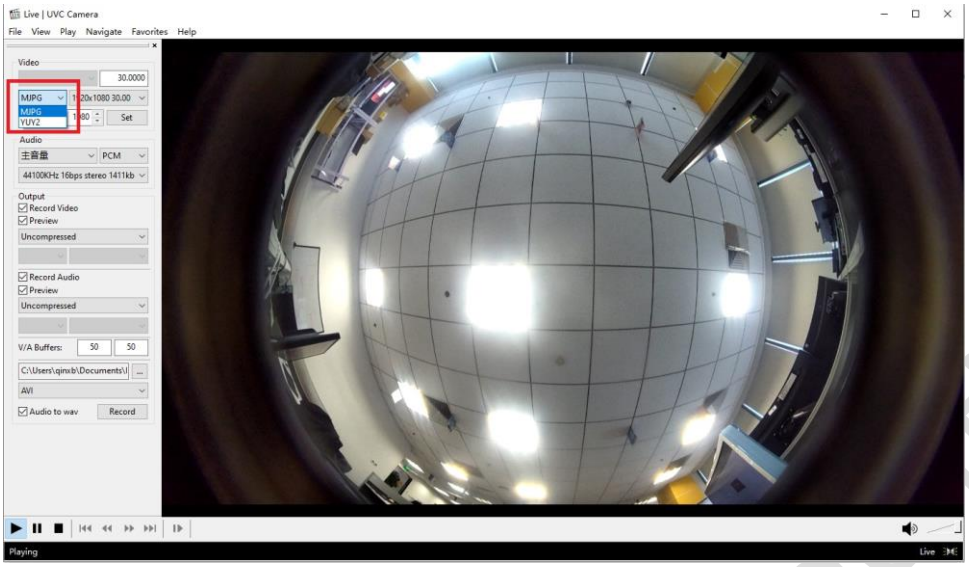


Figure 4-5. Change Format

Step 6.Select the desired resolution.



Figure 4-6. Change Resolution

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Step 7.Continue your operations with the **Play**, **Pause**, and **Stop** buttons as required.

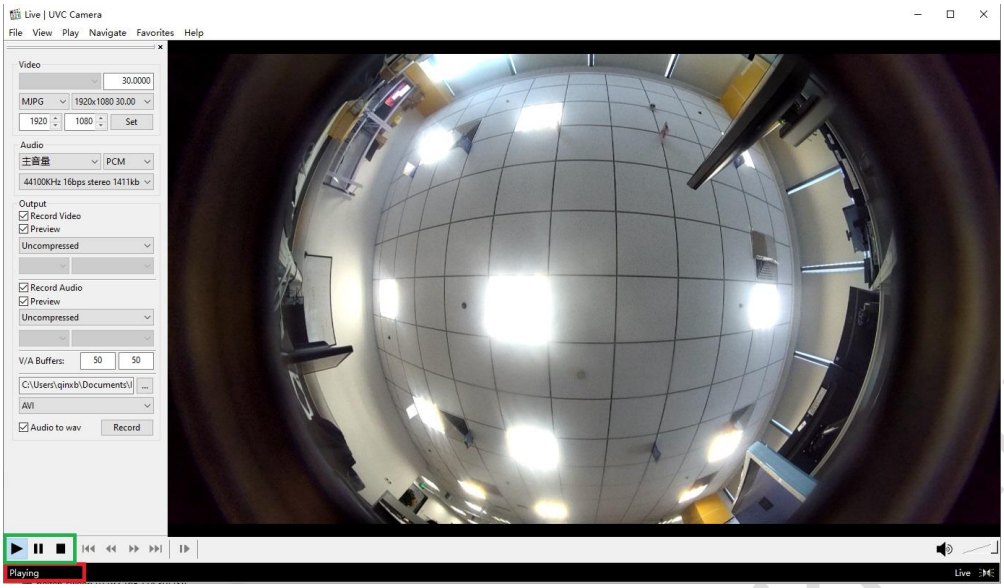


Figure 4-7.Test Operations

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Chapter 5. Close UVC and UAC Streams

To close the UVC and UAC streams, you can click the close button located in the upper right corner of the MPC-HC screen.

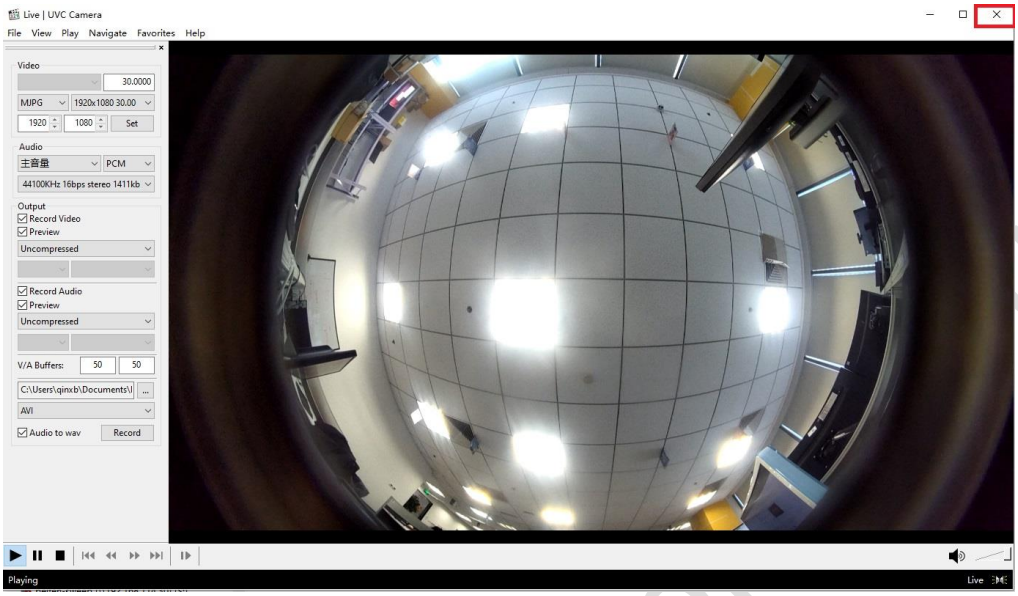


Figure 5-1. Close UVC and UAC Streams

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Chapter 6. Reference Document

C610_Open_Kit_UVC_UAC_Device_User_Manual_(for_LA1.0).pdf

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